Chapter

11 Synthesis

1. **[9 marks]** *6.2.8 (2015:33)*

Soaps and detergents are organic chemicals used to clean greasy material from surfaces.

(a) Draw the general structure of a typical detergent formula unit. [2]

(b) Explain why detergents are soluble in both water and grease. [6]

(c) State why detergents are more effective in hard water than soaps. [1]

___

___

___

2. **[6 marks]** *6.2.5, 6.2.6 (2016 SP:37)*

Ethanol may be produced by fermentation or the hydration of ethene. Conditions are indicated in the table below.

	Temperature (°C)	**Pressure (kPa)**	**Raw materials**
fermentation	60	101.3	
hydration of ethene	300	7000	

(a) Complete the table above to indicate the raw materials for each process. [2]

(b) Explain the lower temperature conditions of the fermentation process. [2]

___

___

___

___

(c) In addition to lower temperature conditions, state **two** other advantages of the fermentation process compared with the hydration of ethene. [2]

___

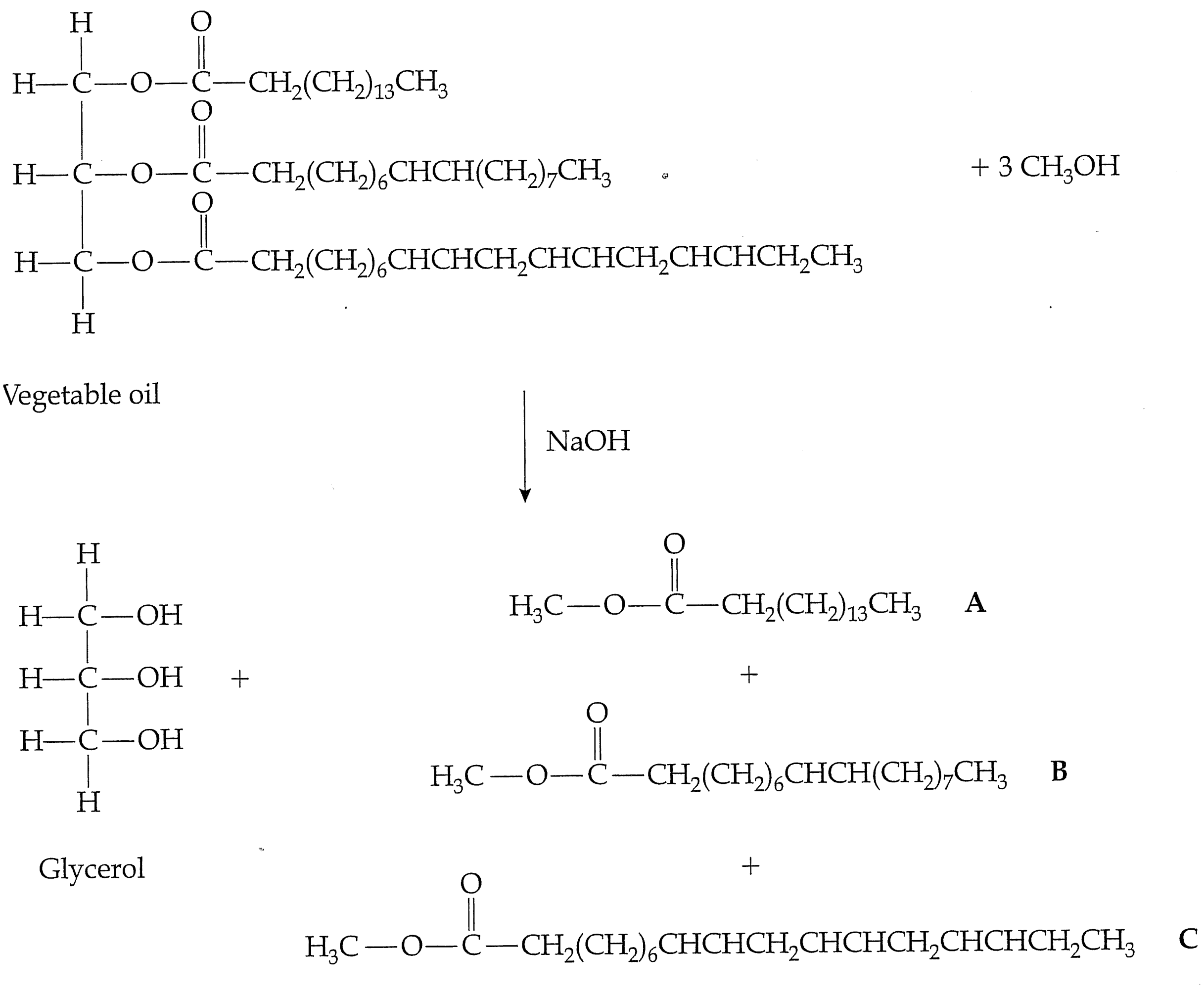
___

___

___

3. **[16 marks]** *6.1, 6.2.4 (2016 SP:40)*

Biodiesel can be produced by a trans-esterification reaction between vegetable oil and an alcohol in the presence of sodium hydroxide catalyst. A typical trans-esterification reaction is shown below. The products are glycerol and three methyl esters.



Methyl esters

(a) Name another catalyst that can be used in the production of biodiesel. [1]

____

(b) The vegetable oil in the reaction has a molar mass of 855.334 g mol^{-1}. If 1.50 tonnes of vegetable oil is reacted, what mass of methanol will be required to react with this amount of oil?
(1 tonne = 1×10^6 g) [3]

(c) Three different methyl esters, denoted by **A**, **B** and **C**, are produced from this reaction. What is the mass of Ester **A** produced in this process if the reaction is 78% efficient in production of this ester? Express your final answer to the appropriate number of significant figures. [5]

(d) The glycerol produced can be used as anti-freeze due to its high water solubility. Explain, with the aid of a diagram, why glycerol is soluble in water [5]

(e) To prevent different products forming in an alternative synthesis pathway, the quantity of sodium hydroxide present in the reaction must be kept low, compared with the vegetable oil. If the mole ratio of NaOH to vegetable oil approaches the ratio 3:1, the alternative pathway becomes significant.

(i) What type of organic product forms in this alternative pathway? [1]

(ii) Draw the structure for **one** organic product that forms in the alternative synthesis pathway from this vegetable oil. [1]

4. **[6 marks]** *6.2.4 (2016:28)*

While petroleum diesel and biodiesel are produced differently, they have similar structures to each other.

(a) The condensed structure of a petroleum diesel is given here.

$$CH_3 - CH_2 - (CH_2)_{12} - CH_2 - CH_3$$

Draw the condensed structure of a biodiesel containing the same number of carbon atoms in the chain. [3]

(b) Biodiesel can be synthesised using a base-catalysed method or a lipase-catalysed method. Outline briefly an argument to justify the use of a lipase-catalysed method rather than a base-catalysed method to produce biodiesel. [3]

5. **[5 marks]** *6.2.6 (2016:30)*

The process of chemical synthesis may involve a sequence of reactions.

(a) Use equations to show how ethyl ethanoate can be produced from ethene through the successive processes of hydration and esterification. [4]

(b) Write the overall equation for the process of synthesising ethyl ethanoate from ethene. [1]

6. **[10 marks]** *6.2.2 (2017:29)*

In a beaker 12.00 mL of 0.0334 mol L^{-1} sulfuric acid solution, $H_2SO_4(aq)$, is added to 32.50 mL of 0.0288 mol L^{-1} potassium hydroxide solution, $KOH(aq)$.

(a) Identify the limiting reagent in this reaction. Show all workings. [5]

(b) Calculate the final concentration of the excess reagent. Show all workings. [3]

(c) Calculate the pH of the final solution. Show all workings. [2]

7. **[8 marks]** *6.2.4 (2017:30)*

Carbonyl chloride, $COC\ell_2$, is a colourless, poisonous gas that is used in the production of insecticides and a variety of plastics. It is produced through the exothermic reaction between carbon monoxide and chlorine gases. Carbonyl chloride is a liquid below 8 °C at 100.0 kPa.

The following equation is used to represent the reaction.

$$CO(g) + C\ell_2(g) \rightleftharpoons COC\ell_2(g) \qquad \Delta H < 0$$

(a) For this industrial process state the conditions that would optimise the: [2]

- rate of reaction

- yield

(b) State one compromise in conditions that might be required to produce carbonyl chloride, $COC\ell_2$, in an industrial process. Explain the effect of this condition and justify why this compromise is required. [6]

6.2.1, 6.2.2 (2017:36)

8. **[16 marks]**

Australia is a significant producer of antimony. Antimony, Sb, and its compounds have a wide range of uses. The metal is used to form alloys with other metals, such as lead, to increase their hardness, while compounds of antimony can be used in the manufacture of many substances such as plastics, pigments and match heads.

High-grade antimony ores are converted to the metal through the use of a blast furnace.
- Antimony sulfide ore is first heated to convert it to an oxide.
- Antimony oxide is then heated with carbon to convert it to a metal.

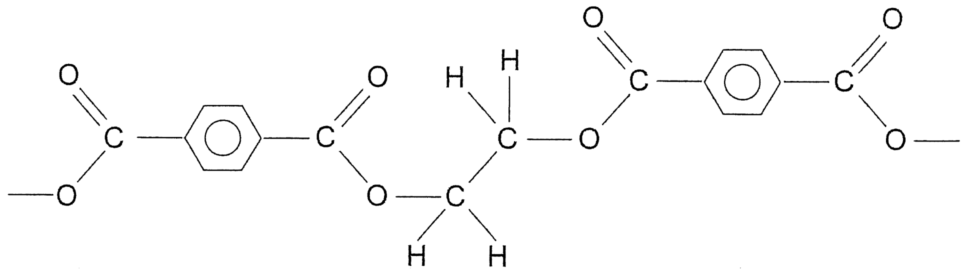
The following equations represent these two reactions.

Reaction one: $2\ Sb_2S_3(s) + 9\ O_2(g) \rightleftharpoons 2\ Sb_2O_3(s) + 6\ SO_2(g)$
Reaction two: $2\ Sb_2O_3(s) + 3\ C(s) \rightleftharpoons 4\ Sb(s) + 3\ CO_2(g)$

(a) What mass of ore would be required to produce 6.00 tonnes of antimony, assuming the ore contains 25.6% by mass of antimony(III) sulfide and the reactions go to completion? [6]

(b) Calculate the maximum volume of sulfur dioxide that could be produced in Reaction one at 525.0 °C and 105 kPa. Give the answer to the correct number of significant figures. [4]

Pure antimony(III) oxide is used as a catalyst in the production of polyethylene terephthalate (PET).



A section of a PET polymer

(c) Draw the monomers required to produce this polymer. [4]

PET is produced through condensation polymerisation; another type of polymer is produced through addition polymerisation. Each of these types of polymerisation uses different types of monomers.

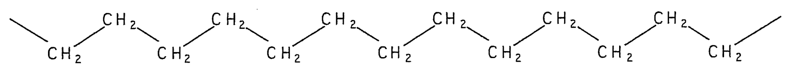
(d) Distinguish between the types of monomers used for each type of polymerisation. [2]

9. **[25 marks]** *6.2.7, 6.2.8, 6.2.9 (2017:39)*

A cosmetic company advertises a range of 'inspiring quality organic, natural and essential personal care ingredients' in its skin care, hair care, aromatherapy and soap products. It claims that the soaps it sells are made from different ingredients boasting 'an array of perfumes and cosmetic benefits'.

Soaps are a class of substances used to clean grease, dirt or oils from a surface such as skin. They do this because they are capable of dissolving in both aqueous and oily systems at the same time.

(a) (i) On the diagram below:
- complete the structure of a soap
- identify and label the key structural features of soap
- draw two molecules of water showing how they are orientated about soap. [5]



The process of dissolving is a consequence of attractive forces between solvent and solute. The different parts of soap are capable of producing different types of attractive forces.

(ii) Name and explain the origin of the predominant attractive force exhibited between the composite particles of soap and water. [3]

(iii) Name and explain the origin of the predominant attractive force exhibited between the composite particles of soap and oil. [3]

(b) Explain why soaps do not function very effectively in hard water. [2]

____

____

____

____

Fats and oils are essentially esters of fatty acids. These esters are called 'triglycerides' and are derived from glycerol and three fatty acids.

(c) (i) Name the functional group in glycerol. [1]

____

(ii) State the two distinctive parts of a fatty acid used to make soap. [2]

One: ____

Two: ____

Below is a typical animal fat (triglyceride).

$CH_2OOC(CH_2)_{14}CH_3$
|
$CHOOC(CH_2)_{16}CH_3$
|
$CH_2OOC(CH_2)_7CH = CH(CH_2)_7CH_3$

To produce soap, the above fat can be hydrolysed with concentrated sodium hydroxide solution.

(d) Draw structural formulae of the four products from this saponification process. Names are not required. [4]

(e) Why are soap solutions basic? [2]

Under Australian law, any company wishing to make soap commercially using a saponification process must register with the National Industrial Chemicals Notification and Assessment Scheme (NICNAS) administered by the Department of Health.

(f) State one health risk caused by chemicals used in the saponification process that would require careful monitoring by NICNAS. [1]

The following table claims to list soaps in increasing order of cleaning effectiveness.

Soaps and their chemical structure

Common name	**Chemical structure**
Sodium caprylate	$CH_3(CH_2)_6COONa$
Sodium caprate	$CH_3(CH_2)_8COONa$
Sodium laurate	$CH_3(CH_2)_{10}COONa$
Sodium myristate	$CH_3(CH_2)_{12}COONa$
Sodium palmitate	$CH_3(CH_2)_{14}COONa$
Sodium stearate	$CH_3(CH_2)_{16}COONa$
Sodium arachidate	$CH_3(CH_2)_{18}COONa$
Sodium behenate	$CH_3(CH_2)_{20}COONa$
Sodium lignocerate	$CH_3(CH_2)_{22}COONa$
Sodium cerotic	$CH_3(CH_2)_{24}COONa$

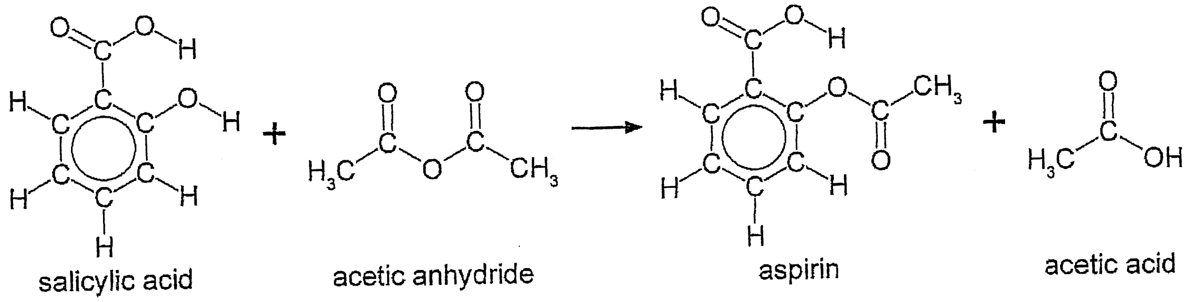
least effective

most effective

(g) Use the information in the table to write an hypothesis that could be used to investigate cleaning effectiveness. [2]

10. [17 marks] *6.2.2, 6.2.3 (2018:40)*

Acetylsalicylic acid is better known as aspirin. It is used to treat pain and inflammation. Aspirin can be synthesised from salicylic acid and acetic anhydride ($C_4H_6O_3$). This process can be represented by the equation below.



The molar mass of salicylic acid is 138.121 g mol^{-1}.
The molar mass of aspirin is 180.158 g mol^{-1}.
This reaction is an equilibrium reaction with a K value of approximately 5.

(a) In the synthesis of aspirin, 45.0 g of salicylic acid was reacted with excess acetic anhydride. This produced 50.2 g of aspirin. What was the percentage yield of this reaction? [4]

____

____

____

____

____

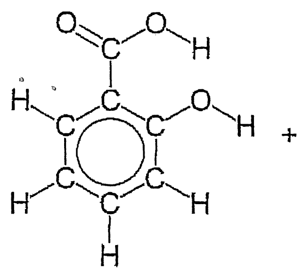
____

____

____

A student conducts a titration to determine the percentage purity of a sample of salicylic acid that was to be used in the synthesis of aspirin.

(b) Complete the equation for the reaction between salicylic acid and sodium hydroxide solution. [2]



An amount of 3.55 g of salicylic acid was dissolved in distilled water and added to a 250.0 mL volumetric flask, which was filled to the calibration line with additional distilled water. 20.00 mL aliquots of the acid solution were titrated against 0.0966 mol L^{-1} sodium hydroxide solution. The average titre obtained was 18.45 mL.

(c) Calculate the mass of salicylic acid in the sample and therefore the percentage purity of the sample. [5]

During the titration, the student used the following procedure.

Number	Procedure
1	Swirl the conical flasks while adding sodium hydroxide solution from the burette.
2	Use the same number of drops of indicator for each titration.
3	Stop the titration at the first sign of the indicator showing a colour change.
4	Wash the pipette with distilled water before filling with salicylic acid solution.
5	Slow down the addition of sodium hydroxide solution as the end point is approached.
6	Rinse down the sides of the conical flask during the titration.

(d) Identify **two** incorrect procedures from the list above, select the effect on the calculated concentration of salicylic acid and give the reason for the effect. [6]

Number	**Effect on calculated concentration** (circle your answer)	Reason
	increase decrease no change	
	increase decrease no change	

11. **[8 marks]** *6.2.1, 6.2.4 (2019:29)*

Sulfuric acid is a very useful chemical that is produced industrially by a multi-stepped process. These steps are summarised by the following equations.

Equation 1 $S(\ell) + O_2(g) \rightarrow SO_2(g)$

Equation 2 $2\ SO_2(g) + O_2(g) \rightleftharpoons 2\ SO_3(g) + 198\ kJ$

Equation 3 $H_2SO_4(\ell) + SO_3(g) \rightarrow H_2S_2O_7(\ell)$

Equation 4 $H_2O(\ell) + H_2S_2O_7(\ell) \rightarrow 2\ H_2SO_4(\ell)$

When dihydrogen sulfate, $H_2SO_4(\ell)$, is mixed with water, it produces sulfuric acid, $H_2SO_4(aq)$.

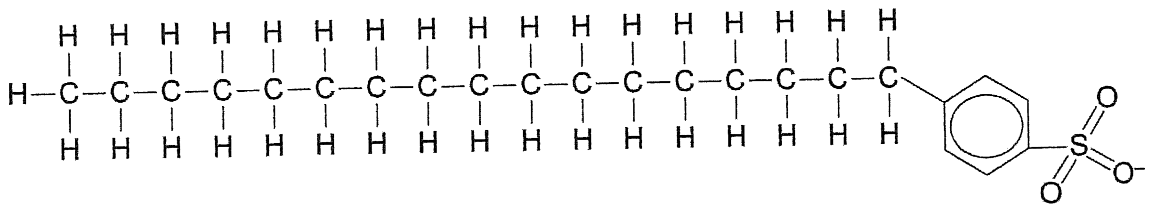
(a) Combine these equations to produce an overall equation for the production of dihydrogen sulfate, $H_2SO_4(\ell)$, from sulfur dioxide, $SO_2(g)$. [2]

(b) Complete the following table by listing the advantages and disadvantages of using high temperatures and high pressures for the reaction represented by Equation 2 above. Consider yield, rate, cost and safety. [6]

	Advantage/s	**Disadvantage/s**
High temperature		
High pressure		

12. **[24 marks]** *6.2.7, 6.2.8, 6.2.9 (2019:37)*

Detergents and soaps are both used as cleaning agents. The general structure of a detergent is given below.



(a) Explain how detergents are able to remove grease from a surface by referring to the intermolecular forces present. Include a labelled diagram to illustrate your answer. [7]

Detergents are considered to be more versatile cleaners than soap.

(b) Explain why soaps are generally less effective than detergents as cleaning agents in hard water. Include a relevant equation in your answer. [4]

Alkenes can also form soaps.

(c) Draw a structural diagram for the soap ion, $C_{17}H_{31}CO_2^-$ using the incomplete structure below. Show **all** atoms and bonds. [2]

C—C—C—C—C—C—C—C—C—C—C—C—C—C—C—C—C

(d) Write an equation showing the formation of this soap from the fat (triglyceride) shown below. [3]

$$\begin{array}{l} C_{17}H_{31}COOCH_2 \\ \qquad\qquad\quad | \\ C_{17}H_{31}COOCH \quad + \\ \qquad\qquad\quad | \\ C_{17}H_{31}COOCH_2 \end{array}$$

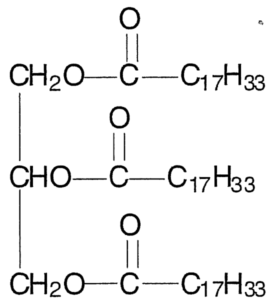
The formation of soap is both an endothermic and equilibrium reaction.

(e) Predict and explain the conditions that would result in the highest yield of soap in the shortest amount of time. [8]

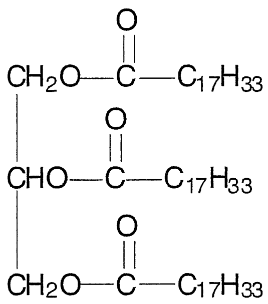
13. **[16 marks]** *6.1 (2020:40)*

Thousands of fast-food outlets across Australia use vegetable oil in cooking. Large volumes of vegetable oil waste are thus produced and need to be disposed of. A disposal option is turning the vegetable oil waste into biodiesel.

Vegetable oil waste is a mixture of free fatty acids and triglycerides. Triolein, the triglyceride of the free fatty acid oleic acid, is typically present in large amounts. The condensed structural formulae of oleic acid and triolein are shown below.

$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	
Oleic acid	Triolein (the triglyceride of oleic acid)

(a) Write a balanced equation, using condensed structural formulae, to show the formation of biodiesel from triolein and ethanol. Assume that a suitable catalyst is present. [3]



(b) Lipase is a protein that can be used to catalyse the reaction between triolein and ethanol. To which class of biological chemicals (other than proteins) does lipase belong? [1]

___

The free fatty acids found in vegetable oil waste will react with the ethanol that was intended for biodiesel synthesis, establishing an equilibrium.

(c) Complete the following equation to show the equilibrium that is established between oleic acid and ethanol. Represent all organic substances as condensed structural formulae and assume acidic conditions. [2]

$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH + CH_3CH_2OH$

$\rightleftharpoons$

In an industrial setting, reaction conditions are adjusted to favour the forward direction of the oleic acid/ethanol equilibrium.

(d) Identify **two** different actions that can be carried out to favour the forward direction of this equilibrium. [2]

One: ______________________________

Two: ______________________________

The base sodium hydroxide can also catalyse the reaction between triolein and ethanol. The free fatty acids in the vegetable oil waste also react with the base.

(e) (i) Write a balanced equation showing the reaction of oleic acid with sodium hydroxide. Represent all organic substances as condensed structural formulae. [2]

$CH_3(CH_2)_7CH=CH(CH_2)_7COOH$

(ii) To which class of compounds does the organic product of this reaction belong? [1]

____

(f) Which of the catalysts, lipase or sodium hydroxide, is more likely to be the industrially preferred catalyst when using vegetable oil waste to make biodiesel? Justify your answer. [3]

____

____

____

____

____

____

(g) Other than the recycling of vegetable oil waste, give **two** different reasons why the production of biodiesel from vegetable oil waste is an example of green chemistry but the production of diesel from fossil fuels is not. Each of your reasons needs to contrast biodiesel and fossil fuel diesel. [2]

One: ______________________________

____

____

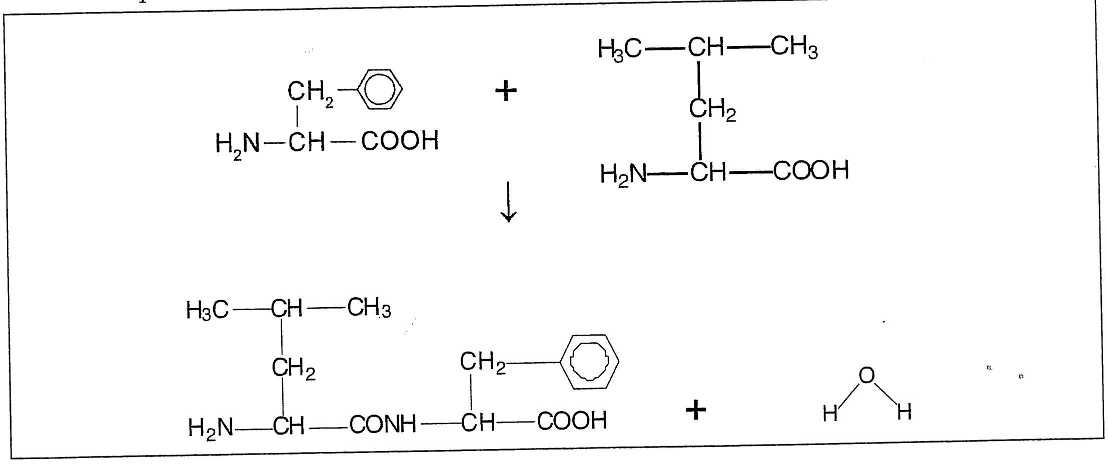
Two: ______________________________

____

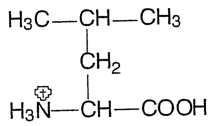
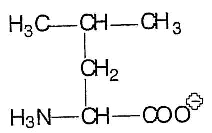
____

d) A definition question - Secondary structure.

e)



f)

Structural formula of leucine	pH
	acidic
	basic

Chapter 11: Synthesis

1.(2015:33) a) $CH_3-(CH_2)n-$⌬$-SO_3^-Na^+$ or $CH_3-(CH_2)n-SO_3-Na^+$

b) The long chain hydrocarbon tail is non-polar so its main force of attraction is through dispersion forces. This end of the detergent is most soluble in non-polar substances such as oil, fats and grease.

The charged end of the detergent is ionic so the predominant forces of attraction are ion-dipole forces. Thus detergent is most soluble in polar substances such as water.

c) Soaps combine with Ca^{2+} and/or Mg^{2+} ions in hard water to form a precipitate while detergents do not form precipitates

2.(2016 SP:37) a)

	Temperature (°C)	Pressure (kPA)	Raw material
fermentation	60	101.3	plant material
hydration of ethene	300	7000	crude oil

b) The process uses enzymes to catalyse reactions at lower temperatures where the ethanol does not boil off and the enzymes do not die. The reaction procures economic rates at lower temperatures lower than industrial catalysts

c) The process uses renewable raw materials, has lower emissions, and can be produced from waste material

3.(2016 SP:40) a) lipase

b) n(veg oil) = $1.50 \times 10^6/855.334 = 1.754 \times 10^3$ mol

$n(CH_3OH) = 3 \times$ n(veg oil) $= 5.261 \times 10^3$ mol

$m(CH_3OH) = 5.261 \times 10^3 \times 32.042 = 1.69 \times 10^5$ g

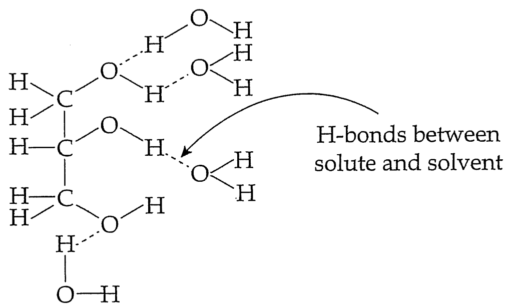
c) for 100% efficient: n(A) = n(Veg oil) = 1.754×10^3 mol

78% efficient, thus n(A) = $0.78 \times 1.754 \times 10^3 = 1.368 \times 10^3$ mol

MF Ester A is $C_{17}H_{34}O_2$ thus M(A) = 270.442 g mol^{-1}

m(A) = $1.368 \times 10^3 \times 270.442 = 3.70 \times 10^5$ g

d) Possible diagram



e) i) soap/sodium salt of long chain fatty acid

ii) sodium salt for long chain fatty acid component of either A, B or C

4.(2016:28) a) Petroleum diesel is a hydrocarbon (alkane) while biodiesel is an ester. You would expect biodiesel to be the methyl ester but with this question you could offer any ester. i.e. offer a methyl, ethyl, or propyl ester but the total number of carbons must be 16.

For example – This is the methyl ester which is normally formed in making biodiesel....

$$CH_3-(CH_2)_{12}-CH_2-\overset{\overset{\displaystyle O}{||}}{C}-O-CH_3$$

b) Using a lipase catalysed method to produce biodiesel is a more 'green' method of chemical synthesis. The specificity of the enzyme allows no side reactions to occur, giving a higher yield of biodiesel and no soap production as seen in base catalysed methods. This means there are no costly separation processes needed to remove the soaps and separation of biodiesel from glycerol is easier. The soap causes the biodiesel to dissolve in the glycerol. This method requires a milder pH temperature and pressure – hence the cost of production is lower and reaction conditions are safer. The use of lipase removes the need to dispose of highly caustic (high pH) waste material.

5.(2016:30)

a)

1: Hydration

ethene to ethanol

$CH_2{=}CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g)$

2: Esterification

ethanol to ethyl ethanoate

$CH_3COOH(g) + CH_3CH_2OH(g) \rightleftharpoons CH_3COOCH_2CH_3(g) + H_2O(g)$

b)

$CH_2{=}CH_2(g) + CH_3COOH(g) \rightleftharpoons CH_3COOCH_2CH_3(g)$

6.(2017:29) a) $H_2SO_4 + 2\ KOH \rightarrow K_2SO_4 + 2\ H_2O$

H_2SO_4 : n = cV = 0.0334 × 0.012 = 4.00×10^{-4} mol

So mol H^+ = 2 × 4.00×10^{-4} = 8.01×10^{-4} mol of H^+

KOH : n = cV = 0.0288 × 0.0325 = 9.39×10^{-4} mol of OH^-

As $H^+ + OH^- \rightarrow H_2O$ there is a 1:1 ratio

The H^+ is the lesser quantity so H_2SO_4 is the limiting reagent

b) the OH^- inxs is $9.36 \times 10^{-4} - 8.016 \times 10^{-4} = 1.344 \times 10^{-4}$ mol

Total volume = 0.012 + 0.0325 = 0.0445 L

So the $[OH^-]$= n ÷ V = 1.344×10^{-4} ÷ (0.0445) = 3.02×10^{-3} mol L^{-1}

c) pOH = –log 3.02×10^{-3} = 2.520

pH = 14 – pOH = 11.48

7.(2017:30) a) rate of reaction : High temperature and high pressure

yield : low temperature and high pressure

b) This question involves the same logic as the Haber Process.

A moderate heat would have to be used as a compromise.

Rate : The reaction rate is maximised at high temperatures because more particles have sufficient energy to overcome the activation energy and the frequency of collisions is raised allowing more successful collisions.

Yield : However at low temperatures the yield is maximised as the forward reaction is exothermic, so lower temperatures favours the exothermic reaction.

Therefore a moderate temperature is a compromise because it allows the reaction rate to be reasonable along with yield being reasonable, making the process more economical by balancing rate and yield.

8.(2017:36) a) m(Sb) = 6×10^6 g so n(Sb) = m/M = $6 \times 10^6/121.8$ = 49261 mol

There is a 2:4 ratio of Sb_2O_3 : Sb so n(Sb_2O_3) = n(Sb) × 0.5 = 24,631 mol

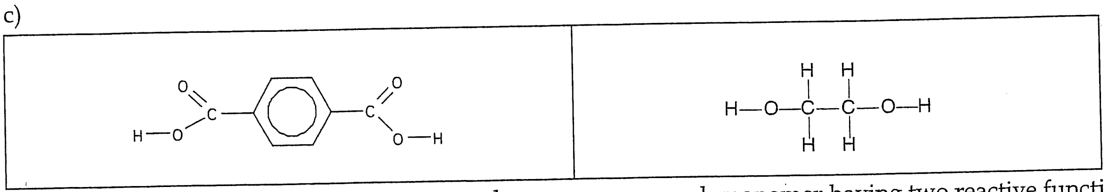
There is a 1:1 ratio of Sb_2O_3 : Sb_2S_3 so n(Sb_2S_3) = 24,631 mol

M(Sb_2S_3) = 339.81 g mol^{-1}

∴ m(Sb_2S_3) = n × M = 8.37×10^6 g

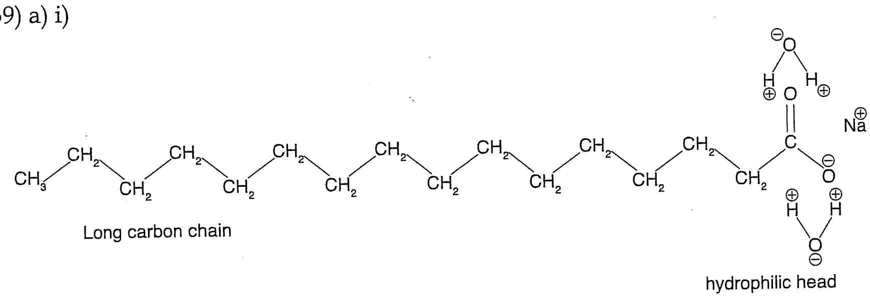
∴ m(ore) = 100/25.6 × 8.37×10^6 = 3.27×10^7 g = 32.7 tonne of ore

b) T = 798.15 K   P = 105 kPa   R = 8.314 J K^{-1} mol^{-1} n = 24631 mol
$n(SO_2)$ = 3/2 × n(Sb) = 3/2 × 49261 = 73891 mol
PV = nRT so V = nRT/P = 73891 × 8.314 × 798.15/105 = 4669783 = 4.67×10^6 L
c)

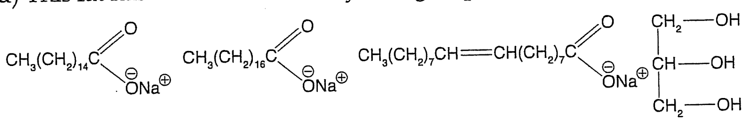


d) Condensation polymerisation requires two base monomers each monomer having two reactive functional groups. Condensation polymers react to eject a small molecule, often water – but can be many other things, so your description must not be exclusively about polyamides and polyesters. Addition monomers involved bonding between alkenes only as the monomer.

9.(2017:39) a) i)



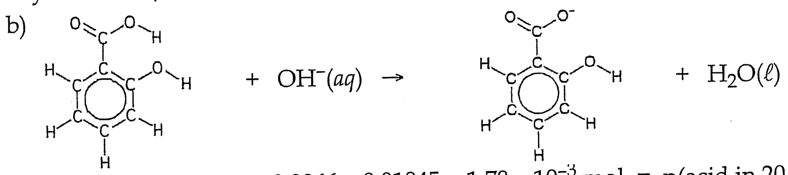
ii) Hydrogen Bonds and Ion-Dipole Bonds:
Hydrogen Bonds: The negative carboxylate ion and the positive hydrogen end of the water dipole are attracted to each other. The OH in water donates a hydrogen bond and the COO^- accepts it
Ion-Dipole Bonds: The Na^+ ion is attracted to the dipole of water and forms an ion-dipole bond.
iii) Dispersion forces: Both oil and soap are large molecules with no dipole changes. Thus they have a large number of electrons and demonstrate instantaneous temporary weak dipoles and attract each other.
b) Stearate ions (soap) precipitate with calcium ions and are no longer available to react with water to form micelles with oil. So the soap is no longer available and does not work.
c) i) Alcohol
ii) A long carbon chain that might have been derived from a fat (hence fatty acid) and a carboxylate group
d) This fat has three different fatty acid groups thus it forms three different soaps and glycerol.



e) Soaps are the sodium salt of a fatty, weak acid. They are a conjugate of a fatty acid. So when they undergo hydrolysis they form the weak acid and the OH^- ion. Thus the number of OH^- ions exceed the H^+ ions and the solution is basic.
f) Sodium hydroxide is 'caustic' which means it dissolves flesh. Care must be taken in its use.
g) As the hydrocarbon chain length increases its effectiveness as a soap increases.

10.(2018:40) a) Percentage yield is = mass obtained ÷ theoretical mass × 100
45.0 g of salicylic acid → 50.2 g of aspirin.
mol(salicylic acid) = 45.0/138.121 = 0.3258 mol salicylic acid
mol(salicylic acid):mol(aspirin) = 1:1 = so 0.3258 mol salicylic acid should → 0.3258 mol aspirin
m(aspirin) = 0.3258 × 180.157 = 58.695 g
but the experiment obtained 50.2 g
% yield = 50.2/58.695 × 100 = 85.5%

b)



c) n (OH^-) in 20.00 mL = 0.0966 × 0.01845 = 1.78×10^{-3} mol = n(acid in 20.00 mL)
n (acid in 250 mL) = 250/20 × 1.78×10^{-3} = 2.23×10^{-2} mol
m (acid) = 138.121 × (2.23×10^{-2}) = 3.08 g
% purity = (3.08/3.55) × 100 = 86.7%
d) Identify **two** incorrect procedures from the list above, select the effect on the calculated concentration of salicylic acid and give the reason for the effect.

Number	3
Effect	decrease
Explanation	there will be a lower titre of NaOH

Number 4
Effect decrease
Explanation the acid will be diluted in the pipette due to residual water from rinsing

11.(2019:29) a) Note the equilibrium arrow and the states and you don't use all four equations – it does not start with S.

$$2\,SO_2(g) + O_2(g) + 2\,H_2O(\ell) \rightleftharpoons 2\,H_2SO_4(\ell)$$

b)

	Advantage	**Disadvantage**
High temperature	Increases rate of reaction	Decreases yield as the reaction is exothermic. Increased energy cost
High pressure	Increases rate of reaction and increases yield as less moles on the right-hand side.	Increased energy cost

12.(2019:37) a) Note this question asks about intermolecular forces present. It does not require hydrophilic and hydrophobic descriptions. You must include a labelled diagram to illustrate your answer.

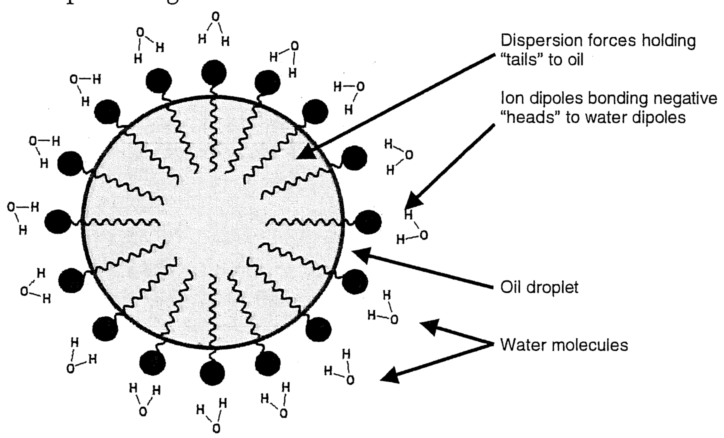
Detergents have a nonpolar dodecyl benzene hydrocarbon chain and a polar sulfonate group that acts as a negatively charged head.

The nonpolar hydrocarbon tail exhibits dispersion forces formed by weak temporary dipoles along the long tail which can interact with and overcome the dispersion forces between the grease particles, thus dissolving into the grease.

The polar sulfonate group is able to interact with the extremely polar water molecules as the negatively charged oxygen is mutually attracted by the positive dipole of the hydrogen in the water–water interactions and forming new ion dipole interactions.

When the detergent particle is dissolved in the grease, micelles form and with agitation the grease will lift from the surface suspended (not dissolved) in the water as micelles and can be washed away.

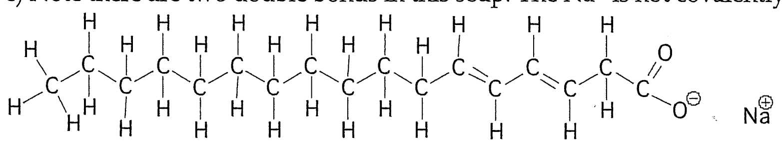
Example of diagram:



b) This question has four parts.

1. Hard water is water containing $Ca^{2+}(aq)$ or $Mg^{2+}(aq)$ ions.
2. Soluble soap (e.g. stearate ion) will precipitate with these ions to form scum but detergents do not.
3. Equation: $Ca^{2+}(aq) + 2\,CH_3(CH_2)_{17}COO^-(aq) \rightarrow Ca(CH_3(CH_2)_{17}COO)_2(s)$
   This does not occur with short chains so a large number of carbons are needed in the equation.
4. This formation of scum means soap is wasted and so cannot clean clothes.

c) Note there are two double bonds in this soap. The Na^+ is not covalently bonded to the soap.



d)

$$\begin{array}{l} H_2COOCC_{17}H_{31} \\ | \\ HCOOCC_{17}H_{31} \\ | \\ H_2COOCC_{17}H_{31} \end{array} + 3\,OH^-(aq) \rightleftharpoons 3\,C_{17}H_{31}COO^-(aq) + \begin{array}{l} CH_2OH \\ | \\ CHOH \\ | \\ CH_2OH \end{array}$$

e) This question is about rate and yield. Think of the five factors of rate to begin.

Rate

High temperature–Rate

High temperature means more energy is available so a higher frequency of collisions as the average kinetic energy of the particles is higher.
More particles have sufficient energy to react when they collide.

High concentration–Rate

High concentration of sodium hydroxide solution so more particles present therefore greater frequency of collisions.
Greater number of successful collisions.

High Surface Area–Agitation–Rate will increase the surface area, increasing the contact /collisions between reacting particles and hence rate.

Yield

High temperature–Yield

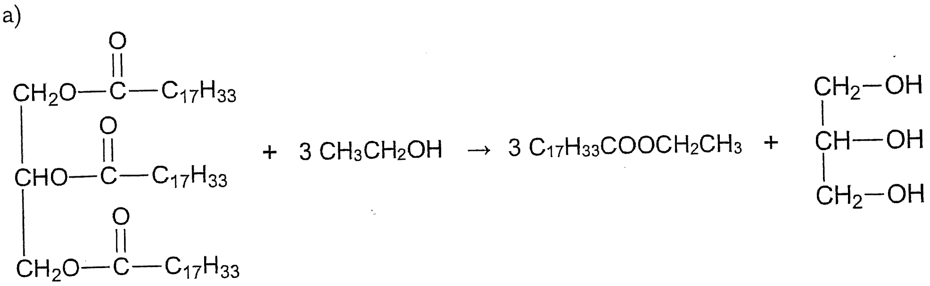
Higher yield as the forward rate will increase more than the reverse rate when temperature is increased as it is endothermic.

Higher Concentration of NaOH(*aq*)–Yield

Higher yield as forward reaction will be faster than reverse reaction until equilibrium re-established.

Removal of soap as it is produced to prevent the reverse reaction – i.e. make it an open system.

13.(2020:40) a)



b) It is an enzyme – an organic catalyst.

c) $CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH + CH_3CH_2OH \rightarrow CH_3(CH_2)_7CH{=}CH(CH_2)_7COOCH_2CH_3 + H_2O$

d) This question is about using Le Châtelier's Principle in a practical situation.

One: Remove product as it forms to move the equilibrium to the product side.

Two: Add reactant to move the equilibrium to the product side.

e) i) This question is about the alkaline hydrolysis of an ester producing the carboxylate ion and water

$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH + OH^- \rightleftharpoons CH_3(CH_2)_7CH{=}CH(CH_2)_7COO^- + H_2O$

ii) The best answer would be soap, but carboxylate ion is correct.

f) Reading the question closely shows 'industrially preferred' rather than 'environmentally preferred'. The answer will concern economy, rate and yield. Interestingly in Australia the manufacture of biodiesel has decreased dramatically in recent years. The world is moving to electric vehicles, not biofuels.

The best answer to this question is sodium hydroxide solution

- can operate at high temperatures and so deliver biodiesel at a high rate
- cheap to purchase
- produce a high yield
- more readily available
- does not denature with pH or temperature changes.

Lipase is also an answer as it does have some advantages including

- no side reactions like sodium hydroxide may have – making soap
- operates at 37 °C and requires less energy
- the catalyst can be recycled
- the catalyst is not dangerous like caustic (dissolves flesh) sodium hydroxide solution

g) There are 12 principles of green chemistry of which you should be aware.

You might select …

- Vegetable oil is renewable while fossil fuel diesel is non-renewable
- Making biodiesel requires less energy than fractional distillation of petro-diesel and therefore less CO_2 is released – you must be clear that this CO_2 is in the <u>making</u> of biodiesel – when they are burned both release comparable quantities of CO_2.
- Biodiesel is carbon neutral – plants absorb CO_2 from the air and release it again when their oils are burned.

Chapter 12: Science Enquiry Skills

1.(2016 SP:38) a) Any two of the following:

- it can be obtained with a high degree of purity and has a known formula
- it undergoes reactions according to known chemical equations
- it is stable (to air)
- it has a high formula mass
- reacts rapidly with acids
- dissolves readily to give standard solutions

b) Calculate the concentration of the standardised HCℓ solution.

$$Na_2CO_3 + 2\,HC\ell \rightarrow CO_2 + H_2O + 2\,NaC\ell$$

$n(Na_2CO_3) = 0.025\ L \times 0.0248\ mol\ L^{-1} = 6.2 \times 10^{-4}\ mol\ Na_2CO_3$

$n(HC\ell) = 2 \times 6.2 \times 10^{-4} = 1.24 \times 10^{-3}\ mol$

$c(HC\ell) = 1.24 \times 10^{-3}\ mol/0.02435\ L = 5.09 \times 10^{-2}\ mol\ L^{-1}$